

PARAMVEER MULTANI

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EDUCATION

McMaster University

Bachelor of Computer Engineering (Co-op)

Hamilton, Ontario

Expected Graduation Apr 2028

- Relevant Coursework: Algorithm Design and Analysis, Data Structures and Algorithms, Software Development, Artificial Intelligence, Advanced Probability and Random Processes, Digital Systems Design, Signals and Systems, Computer Systems Integration, Microprocessors

EXPERIENCE

Hoskin Scientific

Instrument Technician Intern

May 2026 – Sep 2026

Oakville, ON

- Cut calibration report prep from **10 minutes to under 1** by building a browser-based offline tool in **JavaScript** and **ExcelJS** that parses instrument CSV exports into formatted worksheets.
- Drove the tool to **all Hoskin offices across Canada**, where it now backs **20+ calibration reports daily**, by pitching it to management unprompted and shipping it to production.
- Restored **100+** scientific instruments to service by running **PCB-level diagnostics**, isolating failed components through multimeter testing, and resoldering or replacing parts.

BS Transportation

Operations & Technical Assistant

2021 – Present

Hamilton, ON

- Recovered **50+ hours annually** and cut per-manifest time from **5 minutes to under 2** by automating **Avaal** freight manifest generation with a **Python** script.
- Eliminated transcription errors across weekly reporting cycles by rebuilding payroll and expense tracking in **Excel**, replacing a fully manual entry process.

PROJECTS

HousingHub: Student Housing Marketplace

React, TypeScript, Express, Socket.IO, Postgres, Drizzle ORM

Feb 2025 – Present

- Shipped a two-sided housing marketplace serving **24 REST endpoints** on **Express**, covering listing CRUD, image galleries, owner-scoped permissions, bookmarks, and search filters.
- Replaced raw distance with true **multi-modal commute times** (walk, bike, drive, transit) per listing by integrating **Leaflet** with the **OpenRouteService** and **TravelTime** APIs, rendered as route polylines and distance rings.
- Kept types and validation in sync across **9,100+ lines of TypeScript** by modeling a normalized **12-table Postgres** schema in **Drizzle ORM** with **Zod** validators shared by client and server.
- Delivered real-time landlord-student messaging over **Socket.IO** with conversation rooms, read receipts, unread-count tracking, and event-driven notifications.

Tamalife: Cross-Platform Subscription and Warranty Tracker

React Native, Expo, FastAPI, Celery, Firebase, Supabase

Jul 2026

- Owned the end-to-end push notification path for a **React Native/Expo** app by wiring **Firebase Cloud Messaging**, **Supabase** device-token storage, and scheduled **Celery** reminder delivery into deep-linked tap flows.
- Unblocked on-device push after **Expo Go** proved incompatible with native Firebase modules by moving the team to **Expo/EAS** development builds, then validating delivery across physical devices.
- Led UI/UX across **15+ screens** as one of **4 engineers** at **OpenAI Build Week**, building a shared component library and theming system that restyles the whole app from a single palette.

Indoor LIDAR Spatial Mapping System

C, ARM Cortex-M4F, Python

Jan 2026 – Apr 2026

- Reconstructed **3D wireframe models** of indoor spaces from **360°** sweeps by writing embedded **C** on an **ARM Cortex-M4F** to drive a **VL53L1X** ToF sensor over **I2C** and a stepper motor over GPIO.
- Streamed **64 points per revolution** over **UART** at 115200 bps into a **Python/Matplotlib** pipeline that converts polar readings to Cartesian coordinates and renders the scan interactively.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C/C++, SQL, MATLAB, R, LaTeX

Frontend: React, Next.js, React Native, Expo, Tailwind CSS, Radix UI, TanStack Query, Leaflet, Chrome Extensions (Manifest V3)

Backend & Data: FastAPI, Express, Node.js, Socket.IO, WebSockets, PostgreSQL, Drizzle ORM, SQLAlchemy, Redis, Celery, Supabase, Firebase, REST API design

Tools & Infra: Git, GitHub, GitHub Actions, Docker, Linux, pytest, NumPy, PyTorch, Vite, VS Code

Hardware: PCB Diagnostics, Soldering, Multimeter Testing, MSP432E401Y, Arduino, VL53L1X ToF

Interests: Basketball, Chess, Hiking, Hip-Hop, UFC, Catan, Gaming